

**An Explainable Machine Learning Framework for Predictive
Financial Risk Assessment and Early Warning**

PROJECT REVIEW – 2

Under the Guidance of

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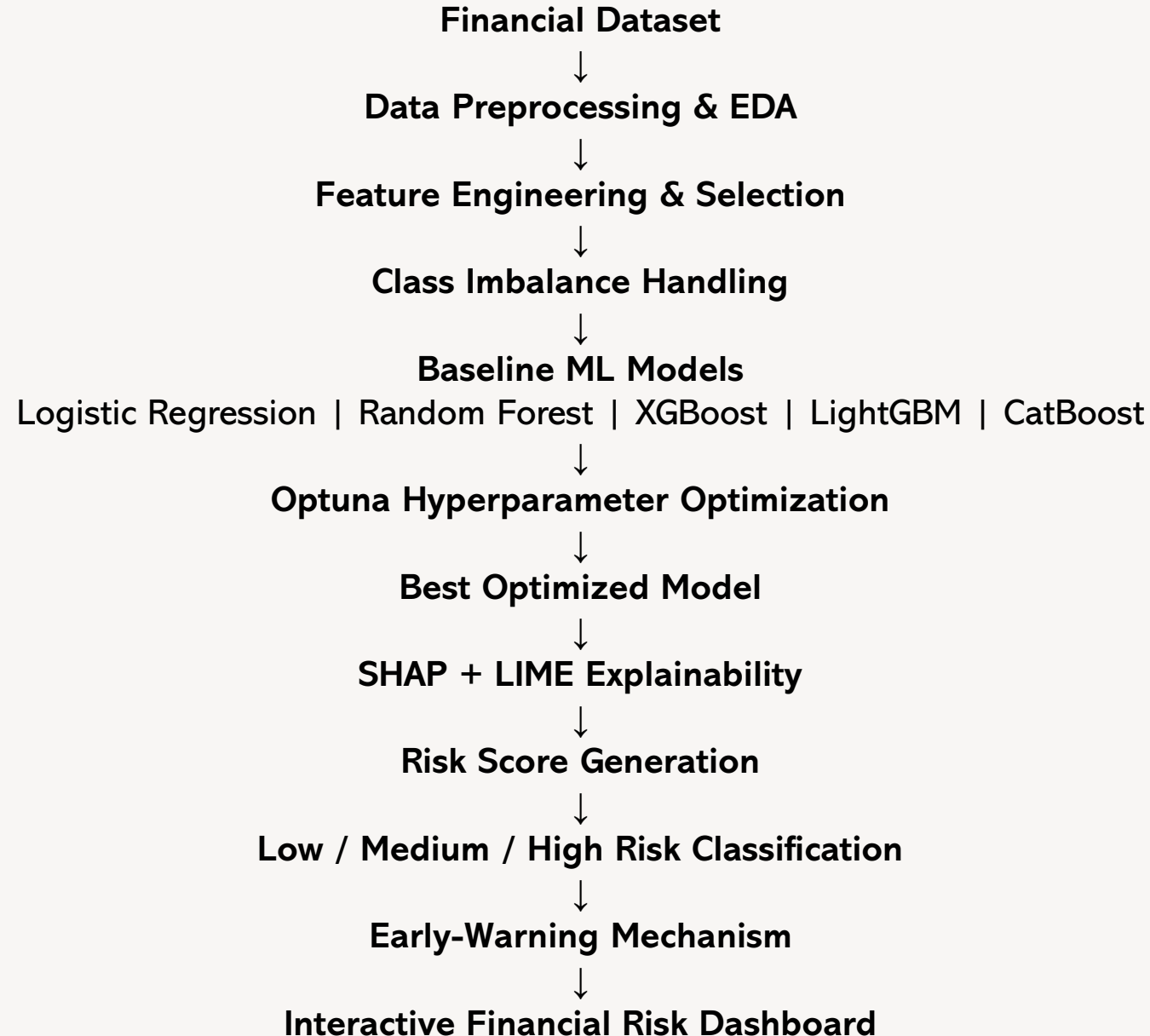
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Contents

1. Proposed Methodology
2. Dataset & Preprocessing
3. System Design / Architecture
4. Baseline Machine Learning Models
5. Optuna-Based Model Optimization
6. Model Evaluation & Selection
7. Explainable AI – SHAP & LIME
8. Risk Score Generation
9. Early-Warning Mechanism
10. Interactive Dashboard
11. Modern Tools & Technologies
12. Implementation & Expected Results
13. Research Paper Draft Progress

Proposed Methodology



Dataset & Data Preprocessing



Dataset Processing

Give Me Some Credit Dataset



Data Cleaning

Handle missing values and duplicates



Exploratory Data Analysis (EDA)

Study feature distributions, correlations, outliers and risk-class distribution



Outlier Handling



Feature Scaling



Feature Engineering & Selection



Class Imbalance Handling

SMOTE / Class Weights



Train-Test Split



Model Training

Final System Output

Default/Risk Probability



Risk Score: 0–100



Risk Category

 Low Risk |  Medium Risk |  High Risk

Dataset: Give Me Some Credit

Purpose:

The dataset will be used to develop a machine-learning model for predicting **customer credit/default risk**.

System Architecture

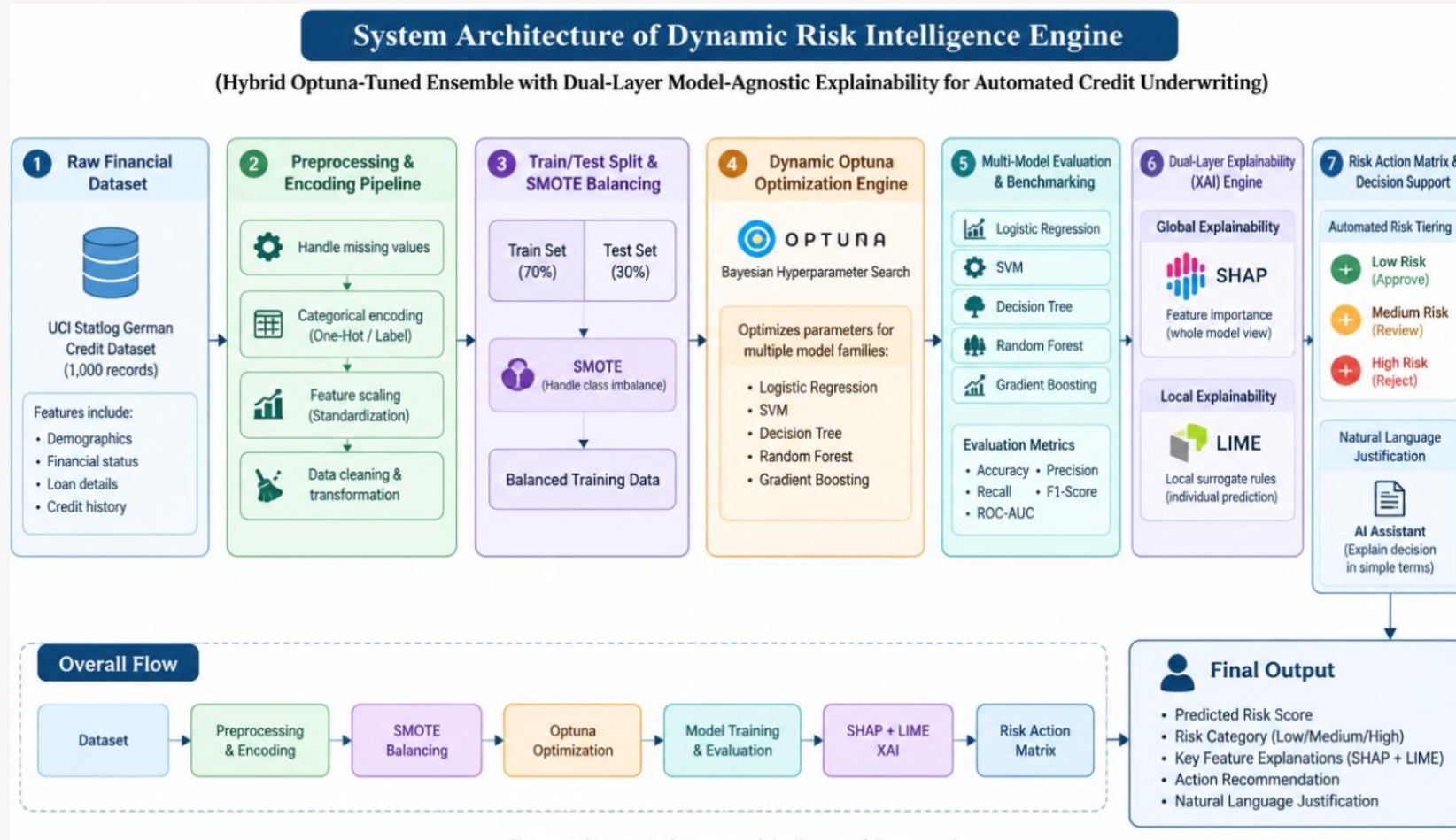


Figure 1: System Architecture of the Proposed Framework

1. **Raw Financial Dataset** – We start with the UCI Statlog German Credit dataset containing demographic, financial, loan, and credit-history features.
2. **Preprocessing** – Missing values are handled, categorical data is encoded, features are standardized, and the data is cleaned.
3. **SMOTE Balancing** – The data is divided into **70% training and 30% testing**. SMOTE is applied to the training data to handle class imbalance.
4. **Optuna Optimization** – Optuna performs hyperparameter optimization for five models: **Logistic Regression, SVM, Decision Tree, Random Forest, and Gradient Boosting**.
5. **Model Evaluation** – The models are trained and compared using **Accuracy, Precision, Recall, F1-Score, and ROC-AUC**.
6. **Explainable AI** – **SHAP** provides global explanations of which features influence the model, while **LIME** explains individual predictions.
7. **Risk Decision Support** – Based on the predicted risk score, the system categorizes applicants into **Low, Medium, or High Risk**, with corresponding actions such as approval, review, or rejection.
8. **Final Output** – The system produces the **risk score, risk category, important feature explanations, recommended action, and a natural-language justification**.

Dataset → Preprocessing → SMOTE → Optuna Optimization → Model Training & Evaluation → SHAP + LIME → Risk Decision → Final Explanation

Baseline Machine Learning Models

Baseline Model Training

Multiple machine-learning models will first be trained using their initial/default configurations.

Model	Purpose
Logistic Regression	Simple baseline model
Random Forest	Handles nonlinear relationships
XGBoost	Boosting-based risk prediction
LightGBM	Efficient gradient boosting
CatBoost	Handles complex/categorical features

Why Multiple Models?

Different ML algorithms may perform differently on financial data.

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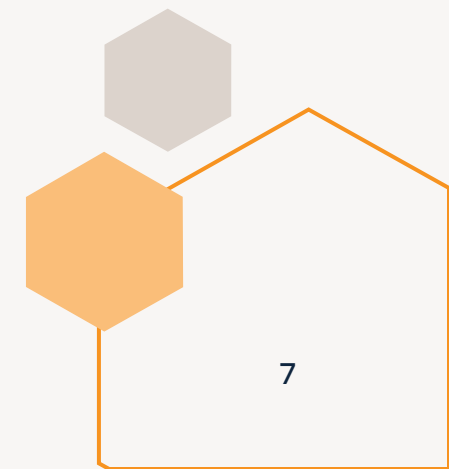
Therefore, we:

Train → Evaluate → Optimize → Compare → Select Best Model

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Evaluation Metrics

Accuracy | Precision | Recall | F1-Score | ROC-AUC | Infidelity | Fidelity | Stability



Optuna-Based Model Optimization

Why Optimization?

Default model parameters may not provide the best prediction performance.

Therefore, we use:

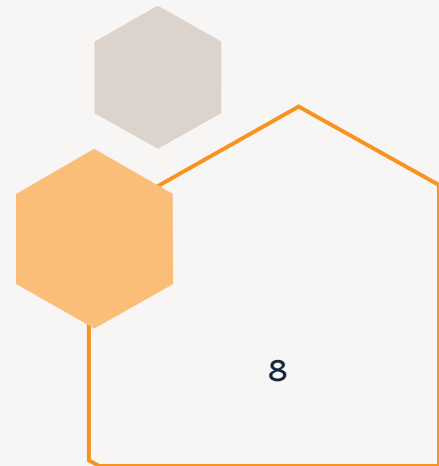
Optuna Hyperparameter Optimization

Optuna automatically searches for better combinations of model hyperparameters.

Example – XGBoost

Optuna can optimize parameters such as:

- `n_estimators`
- `max_depth`
- `learning_rate`
- `subsample`
- `colsample_bytree`



Model Evaluation & Selection

Models Considered

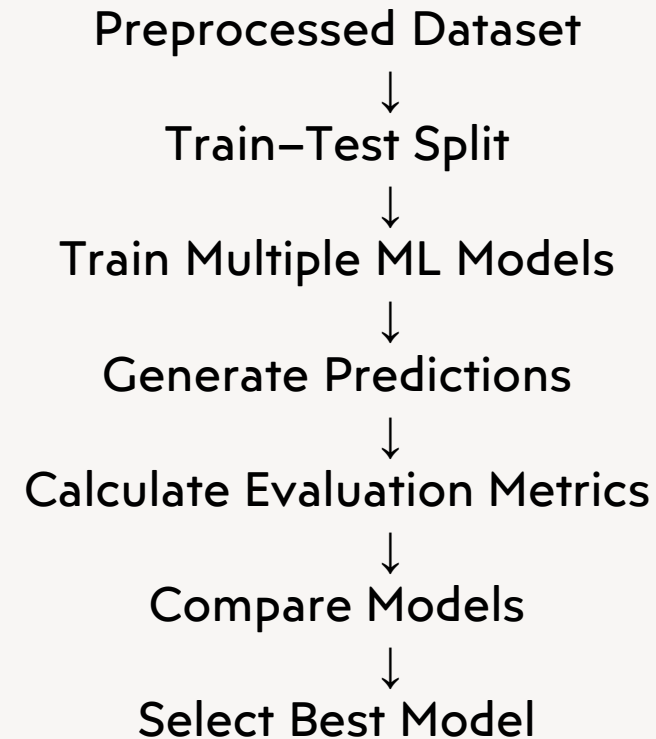
- Logistic Regression
- Random Forest
- XGBoost
- LightGBM
- CatBoost

Evaluation Metrics

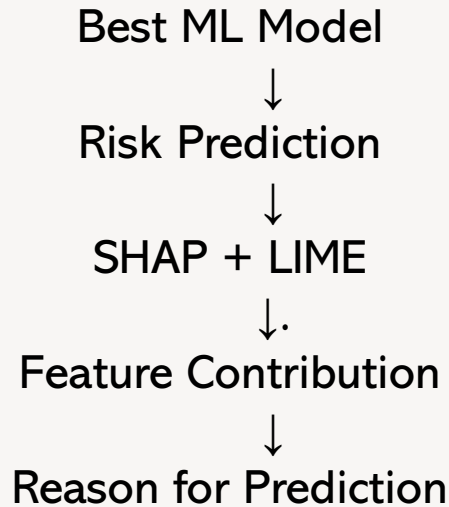
- **Accuracy** – Overall percentage of correct predictions.
- **Precision** – How many predicted risky cases are actually risky.
- **Recall** – How many actual risky cases are correctly detected.
- **F1-Score** – Balance between Precision and Recall.
- **ROC-AUC** – Measures the model's ability to distinguish between risk classes.

Selection

- The model will be selected based on its overall performance across these evaluation metrics, with particular attention to **Recall, F1-Score and ROC-AUC** for risk detection.



Explainable AI – SHAP & LIME



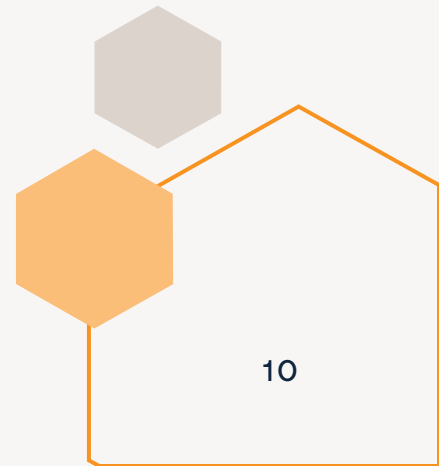
Feature	Contribution
Credit Amount	↑ Risk
Credit Duration	↑ Risk
Credit History	↓ Risk
Savings Account	↓ Risk
Age	↓ Risk

SHAP

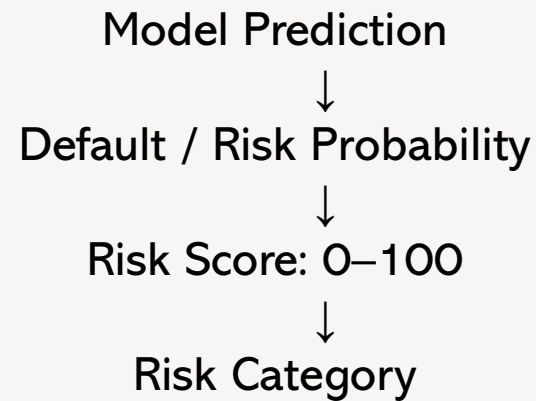
Explains how each feature contributes to a model prediction.

LIME

Explains individual predictions using a local interpretable model.



Risk Score Generation

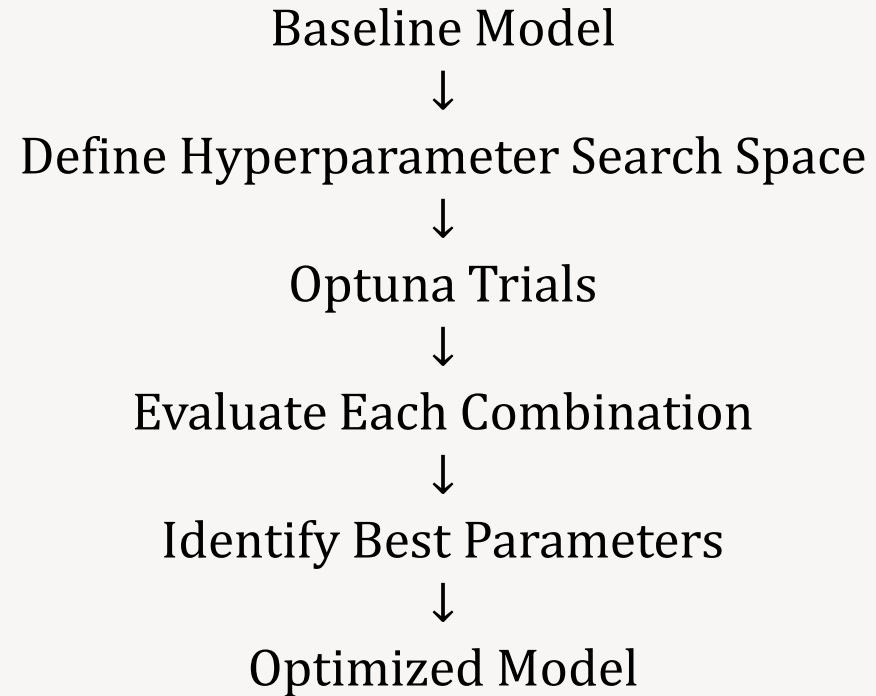


3 types of risk:
Low Risk
Medium Risk
High Risk

Early-Warning Mechanism



Optimization Process



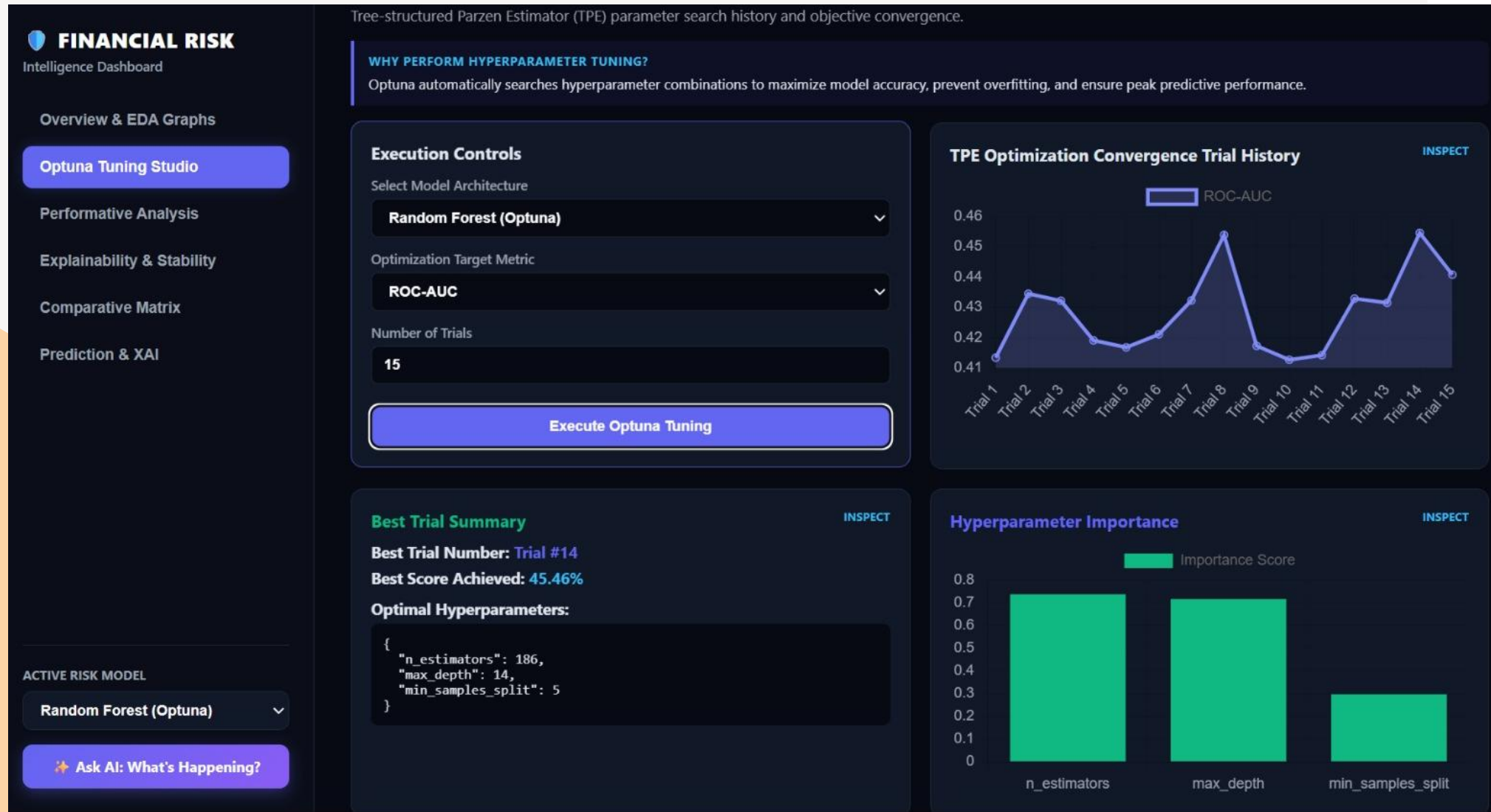
Modern Tools & Technologies

Tool	Purpose
Python	Main programming language
Pandas	Data preprocessing & analysis
Matplotlib	Data visualization
Scikit-learn	ML models & evaluation
XGBoost / LightGBM / CatBoost	Advanced ML models
SHAP + LIME	Explainable AI

Implementation & Expected Results

Dataset Loading	✓
Data Preprocessing	✓
Missing Value Analysis	✓
Duplicate Analysis	✓
EDA	✓
Descriptive Analysis	✓
Outlier Analysis	✓
Feature Engineering	✓
Feature Selection	✓

Interactive Dashboard



Optuna Optimization: The system tested 15 trials and identified the best Random Forest configuration at Trial 14, achieving a 45.46% ROC-AUC for the selected optimization run.





Comparative Results: The models are compared using Accuracy, ROC-AUC, Fidelity, Infidelity, Stability, and Consensus, giving a combined view of both prediction performance and explanation reliability.

THANK YOU

